

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I G.K. Vyshnavi/192425069 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



20/20
25/11

1. analyze whether the problem is primarily caused by the transport protocol or the application layer

The performance issue is caused by both the tcp and the application layer, but the primary cause is the network conditions affecting UDP communication.

Video conferencing applications use UDP because it provide low latency by transmitting packets without acknowledgment or retransmissions. However UDP cannot recover lost packets. The reported 8% packet loss results in missing audio and video data, while 180ms jitter causes packets to arrive at irregular intervals, leading to frozen video and distorted voice.

2. Real time multimedia applications.

Real time multimedia applications have strict Quality of Service requirements:

1. low latency: voice and video should reach users almost instantly
2. low jitter: packets must arrive at regular
3. minimal packet loss: lost packets reduce audio and video quality.

3. Enhance conferencing quality.

1. continue using UDP with RTP for efficient media delivery.
2. implement forward error correction to recover lost packets.

application level improvements:

1. Implement an adaptive jitter buffer that adjusts automatically according to network conditions.
2. Enable adaptive bitrate streaming to reduce video quality during congestion instead of freezing.

Case study 2 - DNS performance improvement.

1. The observed DNS lookup time of 800ms is significantly higher than acceptable values.

Possible reasons include:

1. Centralized DNS server located far from users.
2. Poor DNS caching results in repeated queries.
3. Low TTL configuration causing frequent cache expiration.
4. High network congestion during peak hours.

2. A distributed DNS architecture include:

Anycast DNS:

1. Deploy multiple DNS servers globally using same IP address.
2. Users are automatically directed to the nearest DNS server.

DNS prefetching:

1. Browsers resolve frequently accessed domain names before users request them.
2. This eliminates waiting time during webpage loading.

TTL optimization:

1. configure moderate TTL values (5-30 min)
 2. frequently requested records remain cached while allowing timely updates.
3. Evaluate advantages and disadvantages:

Advantages:

- * Faster DNS resolution
- * Improved scalability for global users.
- * automatic failover during server failures.
- * Reduced latency.

Disadvantages:

- * Higher infrastructure and maintenance cost
- * more complex DNS configuration.
- * anycast requires advanced routing support.
- * incorrect TTL values may delay DNS update.
- * monitoring multiple DNS servers increases administrative effort.

3 Case study 3 - distributed DNS for cloud services

1 DNS architecture on application performance.

A centralized DNS architecture direct all DNS requests to a single primary server.

Impact on performance:

1. creates a single point of failure.
2. High traffic causes server overload.
3. DNS lookup time increases to 800ms.
4. Delays application process.
5. Increased response time reduces customer satisfaction.

2. Design a distributed dns solution that minimizes lookup latency while maintaining high availability

1. multiple authoritative DNS servers across continents.
2. Any cast routing to connect users to the nearest server.
3. Recursive DNS catching at regional locations.
4. Recursive load caching at regional locations.

Benefits:

1. lookup latency below 50ms
2. Higher availability
3. Better scalability.

analyze how TTL values, Any cast routing and DNS caching influence system performance and fault tolerance.

TTL

1. Controls cache duration
2. Reduces repeated DNS queries
3. Improves Response time
4. Excessively long TTL delays record updates

2. Anycast Routing.

1. Routes users to the nearest DNS server.
2. minimizes network latency.
3. provides automatic failure if a server becomes unavailable.

case Study 4 - TCP latency in stock Trading Platform.

Flow control:

TCP flow control prevents the sender from overwhelming the receiver. During heavy trading, reduced receiver window size forces the sender to wait before transmitting additional data, increasing latency.

Nagle's algorithm:

Nagle's algorithm combines multiple small packets before transmission to reduce network overhead. this buffering delays order transmi

SYN Queue limitations:

when thousands of clients simultaneously establish TCP connections, the server's SYN queue becomes full.

2. Recommended Improvements.

1. Disable Nagle's algorithm using `TCP_NODELAY`
2. Increase TCP SYN backlog queue size
3. enable TCP Fast open to reduce connection setup delay.
4. Increase TCP send and receive buffer sizes.
5. use persistent TCP connection
6. optimize retransmission time out.

Expected benefits:

1. lower acknowledgement latency.
2. fast order processing.
3. Improved throughput.
4. Better reliability during peak hours.